

A Simplified Approach to the Basis Functions of Symmetry Operations and Terms of Metal Complexes in an Octahedral Field with d^1 to d^9 Configurations

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Crystal field theory and ligand field theory extensively use the symmetry notations for spectroscopic terms in metal complexes. The terms are derived from group theory, which employs the Mulliken symbols. The three-dimensional irreducible representation is designated with T. However, most textbooks in group theory do not give the definitions of subscripts 1 and 2 of a T representation. Students are left with the impression that the subscript 1 or 2 is arbitrarily assigned. Even though the subscripts 1 and 2 are well-defined in one-dimensional representation A in the octahedral (O_h) point group, there is still no simple explanation as to why a d^3 octahedral complex has a ground state of $^4A_{2g}$ rather than $^4A_{1g}$.

The ambiguity can be clarified with the construction of basis functions as the object of the symmetry operations and the observation of how they are transform with respect to the operations. The irreducible representation can then be assigned. The goal of this article is to demonstrate how to construct the basis sets for d^n metal complexes in an octahedral field.

Discussion

Two examples of ground states are shown for the construction of respective basis functions for symmetry operations. Additional examples are discussed in the supporting information.

d^3 Configuration

If the three d electrons are identified with the numbers 1, 2, and 3, there are six possibilities for these three electrons in three t_{2g} orbitals. The basis function is the linear combination of all six possibilities so that it is antisymmetrical with respect to interchange of any pair of electrons. The first term can be set as $d_{xy}(1)d_{xz}(2)d_{yz}(3)$. This term is arbitrarily set to be positive. The rest of the terms can be generated by interchange of electrons with respect to the first term. A negative sign is given to those terms with one interchange and positive sign for those terms with two interchanges of electrons. Thus, a basis function of

$$\begin{aligned} & d_{xy}(1)d_{xz}(2)d_{yz}(3) - d_{xy}(1)d_{yz}(3)d_{xz}(2) \\ & - d_{xy}(2)d_{xz}(1)d_{yz}(3) + d_{xy}(2)d_{yz}(3)d_{xz}(1) \\ & - d_{xy}(3)d_{xz}(2)d_{yz}(1) + d_{xy}(3)d_{yz}(1)d_{xz}(2) \end{aligned}$$

is obtained. The symmetry operation can be carried out on this basis function to give the symmetry of the function. Symmetry

operations confirm that a d^3 metal complex with O_h symmetry has a ground state of $^4A_{2g}$.

d^5 Configuration (Weak Field–High Spin)

The challenge for a d^5 case is to come up with a unique function to represent the two electrons in the e_g orbital that meets the symmetry criterion. For instance, the C_3 operation will transform $d_{2z^2-x^2-y^2}$ into $d_{2x^2-y^2-z^2}$, which cannot be presented with the original function in any way. The $d_{2z^2-x^2-y^2}$ orbital is actually a linear combination of $d_{y^2-z^2}$ and $d_{z^2-x^2}$, that is, $d_{z^2-x^2} - d_{y^2-z^2}$. The three Cartesian coordinates are identical in an octahedral environment. A function that allows all possible linear combinations of $d_{x^2-y^2}$, $d_{y^2-z^2}$, and $d_{z^2-x^2}$ is needed. If one electron is in $d_{x^2-y^2}$, then $d_{y^2-z^2}$ and $d_{z^2-x^2}$ must be linearly combined to accommodate the second electron. It must be also antisymmetrical to the interchange of the pair of electrons. Hence, the first part of the function is

$$d_{x^2-y^2}(1)d_{y^2-z^2}(2)d_{z^2-x^2}(2) - d_{x^2-y^2}(2)d_{y^2-z^2}(1)d_{z^2-x^2}(1)$$

The combination of $d_{x^2-y^2}$ and $d_{y^2-z^2}$ gives

$$\begin{aligned} & -d_{x^2-y^2}(1)d_{y^2-z^2}(1)d_{z^2-x^2}(2) \\ & + d_{x^2-y^2}(2)d_{y^2-z^2}(2)d_{z^2-x^2}(1) \end{aligned}$$

whereas the combination of $d_{x^2-y^2}$ and $d_{z^2-x^2}$ yields

$$\begin{aligned} & -d_{x^2-y^2}(1)d_{y^2-z^2}(2)d_{z^2-x^2}(1) \\ & + d_{x^2-y^2}(2)d_{y^2-z^2}(1)d_{z^2-x^2}(2) \end{aligned}$$

The function is then

$$\begin{aligned} & d_{x^2-y^2}(1)d_{y^2-z^2}(2)d_{z^2-x^2}(2) - d_{x^2-y^2}(2)d_{y^2-z^2}(1)d_{z^2-x^2}(1) \\ & - d_{x^2-y^2}(1)d_{y^2-z^2}(1)d_{z^2-x^2}(2) \\ & + d_{x^2-y^2}(2)d_{y^2-z^2}(2)d_{z^2-x^2}(1) \\ & - d_{x^2-y^2}(1)d_{y^2-z^2}(2)d_{z^2-x^2}(1) \\ & + d_{x^2-y^2}(2)d_{y^2-z^2}(1)d_{z^2-x^2}(2) \end{aligned}$$

In comparison with the first term, $d_{x^2-y^2}(1)d_{y^2-z^2}(2)d_{z^2-x^2}(2)$, a negative sign is assigned to those terms with the odd number of electron indices change and positive for the even number.

The symmetry can be worked out with this basis function. It is antisymmetrical to the C_2 rotation. The rest of the symmetry operations confirm that the function belongs to A_{2g} symmetry. The three electrons in t_{2g} orbitals have similar symmetry as in the case of d^3 , that is, A_{2g} . Taking the product of A_{2g} and A_{2g} gives rise to a ground-state term of ${}^6A_{1g}$ for the present case.

Conclusion remark

In group theory, the formal procedure of finding the term always starts with the symmetry operation on the basis. "Why does an octahedral Ni(II) (d^8) complex has a ground state of ${}^3A_{2g}$?" is a question without a simple answer. It now can be said

that the two unpaired electrons are evenly spread out through the $d_{x^2-y^2}$, $d_{y^2-z^2}$, and $d_{z^2-x^2}$ orbitals. An A_{2g} term arises from a basis function that is the linear combination of $d_{x^2-y^2}d_{y^2-z^2}d_{z^2-x^2}$.

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Supporting Information Available

An expanded and more detailed version of the article; student handout. This material is available via the Internet at <http://pubs.acs.org>.